

B T G V PRASAD CHATRATHI

Computer Vision Engineer | Machine Learning Engineer | AI Engineer

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PROFESSIONAL SUMMARY

Computer Vision Engineer with 2 years of experience designing and deploying end-to-end AI/ML solutions, from dataset preparation and model training through production deployment. Skilled in building real-time object detection and recognition systems using YOLOv5/YOLOv8, CNNs, and OpenCV, with hands-on experience in transfer learning, hyperparameter tuning, and model optimization for production environments. Delivered a facial recognition and biometric authentication system achieving 90%+ identification accuracy, and an automated safety-monitoring system that cut incident response time by ~70%. Proficient in Python, PyTorch, and deployment tooling including Flask, FastAPI, and Docker. Strong collaborator with cross-functional engineering and product teams, with a track record of taking computer vision solutions from proof-of-concept to production.

PROFESSIONAL EXPERIENCE

Computer Vision Engineer

ImageVision.ai

Nov 2024 - Present

Hyderabad, India

- Engineered an end-to-end computer vision pipeline for the packaging industry - covering data collection, preprocessing, annotation, and validation - producing high-quality training datasets that enabled object detection and product classification models with 98% labeling accuracy.
- Designed, trained, and optimized YOLOv5 and CNN-based object detection models using data augmentation, class balancing, and hyperparameter tuning, improving detection precision, recall, and generalization for real-time and batch inference workloads.
- Built and deployed OpenCV-based computer vision pipelines for real-time inference, reducing model inference latency.
- Collaborated with cross-functional engineering, product, and deployment teams to integrate production-ready CV models into manufacturing workflows, supporting model monitoring, debugging, and continuous improvement in live environments.

PROJECTS

Facial Recognition System - Real-Time Biometric AI System

- Business Problem: Manual identity verification (e.g., attendance, ID checks) is slow and error-prone.
- Solution: Built a real-time facial recognition pipeline using DeepFace with FaceNet-512 embeddings, integrated with iris authentication for a unified biometric verification system.
- Results: Achieved 90%+ identification accuracy across variable lighting, partial occlusion, and diverse demographics; validated as a proof-of-concept for college attendance and government ID verification use cases.

Technologies: Python, OpenCV, Dlib, YOLOv5, CNN, DeepFace, FaceNet-512, PyTorch, Transfer Learning

Unattended Person Detection System for Elevators - Safety AI System

- Business Problem: Vulnerable passengers (children, elderly) left unattended in elevators pose safety risks that manual monitoring cannot scale to address.
- Solution: Real-time person detection and time-occupancy analysis on existing CCTV infrastructure, requiring no additional hardware, with automated multi-channel alerting (audio + WhatsApp).
- Results: Cut average emergency response time by ~70%, improving safety outcomes for vulnerable passengers.

Technologies: Python, YOLOv8, OpenCV, PyTorch, Object Detection, Tracking Algorithms, Real-Time Inference

TECHNICAL SKILLS

Programming Languages: Python

Computer Vision: OpenCV, YOLOv5, YOLOv8, Object Detection, Small Object Detection, Image Classification, Image Segmentation, Facial Recognition, Video Analytics

Machine Learning: Linear Regression, SVM, K-Means Clustering, Neural Networks

Deep Learning: CNN, Transfer Learning, Hyperparameter Tuning, Model Optimization, Real-Time Inference

Deployment: Flask, FastAPI, Streamlit, Docker, REST API

NLP / LLM: LangChain, FAISS, Vector Embeddings, Text Processing, Large Language Models

Data & Visualization: NumPy, Pandas, Tableau, Matplotlib

Soft Skills: Analytical Problem Solving, Critical Thinking, Cross-Functional Collaboration

EDUCATION

Bachelor of Technology (B. Tech), Electronics and Communication Engineering *Feb 2020 - Apr 2024*
Swarnandhra College of Engineering and Technology (Autonomous), Narasapuram, India | GPA: 7.58/10

CERTIFICATIONS

- Natural Language Processing on Google Cloud - Skills: NLP, Text Preprocessing, Vector Embeddings
 - Introduction to Computer Vision and Image Processing (IBM) - Skills: OpenCV, Image Processing, Object Detection, Image Segmentation, Pose Estimation
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VOLUNTEERING

- Team Lead, National Service Scheme, Narasapuram - Aug 2023